

Modelling Cellular Kinematics in Self-Similar Plant Growth

DP Baig

1 Introduction

The diffusion-advection-reaction equation is

$$\frac{\partial U}{\partial t} + as \frac{\partial U}{\partial s} = D \nabla^2 U + p - gU \quad (1.1)$$

where $U = U(\mathbf{s}, t)$ is the concentration, which varies with the position $\mathbf{s}$ and time t ; a is the velocity coefficient; D is the diffusion coefficient; p is the source; and g is the degradation rate.

The parabolic map $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is written as

$$\begin{aligned} s &\mapsto x = sr \\ r &\mapsto y = \frac{1}{2}(r^2 - s^2) \end{aligned} \quad (1.2)$$

the Jacobian of which is

$$J = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial s} \end{pmatrix} = \begin{pmatrix} s & r \\ r & -s \end{pmatrix}. \quad (1.3)$$

The metric is then

$$g = J^T J = \begin{pmatrix} s^2 + r^2 & 0 \\ 0 & s^2 + r^2 \end{pmatrix}. \quad (1.4)$$

The Laplacian is then

$$\begin{aligned} \nabla^2 U &= \frac{1}{\sqrt{g}} \partial_i (\sqrt{g} g^{ij} \partial_j U) \\ &= \frac{1}{s^2 + r^2} (\partial_s^2 + \partial_r^2) U \quad \because \sqrt{\det g} = s^2 + r^2. \end{aligned} \quad (1.5)$$

In general if $g_{ij} = h_j^2 \delta_{ij}$ we have $\sqrt{\det g} = \prod_j h_j$. So the Laplacian in 2D is now

$$\nabla^2 U = \frac{1}{h_1 h_2} \left(\partial_1 \left(\frac{h_2}{h_1} \partial_1 U \right) + \partial_2 \left(\frac{h_1}{h_2} \partial_2 U \right) \right) \quad (1.6)$$

To make the calculation easier we can immediately move to a frame of reference that follows the growth of the plant with the change of variables

$$s = \xi e^{at} \quad (1.7)$$

motivated from the velocity relation $\dot{s} = as$. We will then be interested in transforming the equation from $U(s, r, t)$ to $V(\xi, r, t)$. Using the chain rule

$$\begin{aligned} \frac{\partial U}{\partial t} &= \frac{\partial x^j}{\partial t} \partial_j V = \partial_t V - a\xi \partial_\xi V \\ \frac{\partial U}{\partial s} &= e^{-at} \frac{\partial V}{\partial \xi} \\ \frac{\partial^2 U}{\partial s^2} &= e^{-2at} \frac{\partial^2 V}{\partial \xi^2} \end{aligned} \quad (1.8)$$

So Equation (1.1) becomes

$$\frac{\partial V}{\partial t} = \frac{D}{h_s h_r} \left(e^{-2at} \partial_\xi \left(\frac{h_r}{h_s} \partial_\xi V \right) + \partial_r \left(\frac{h_s}{h_r} \partial_r V \right) \right) + p - gV. \quad (1.9)$$

2 1D

In 1D we have

$$\frac{\partial V}{\partial t} = \frac{D}{h^2(\xi)} e^{-2at} \partial_\xi^2 V + p - gV. \quad (2.1)$$

If we assume that $h(\xi)$ takes the form $h(\xi) = s = \xi e^{at}$ then we have

$$\frac{\partial V}{\partial t} = \frac{D}{\xi^2} e^{-4at} \partial_\xi^2 V + p - gV. \quad (2.2)$$

Looking at the homogeneous part and assuming the $V(\xi, t)$ is separable we can write $V(\xi, t) = \Xi(\xi)T(t)$ and find

$$e^{4at} \left(\frac{\partial_t T}{T} + g \right) = \frac{D}{\xi^2} \frac{\partial_\xi^2 \Xi}{\Xi} = \lambda, \quad (2.3)$$

where λ is some constant. The $T(t)$ part becomes

$$\begin{aligned} \int_{T_0}^T \frac{dT}{T} &= \int_{t_0}^t dt (\lambda e^{-4at} - g) \\ \ln \frac{T}{T_0} &= -\frac{\lambda}{4a} (e^{-4at} - e^{-4at_0}) - g(t - t_0) \\ T(t) &= T_0 e^{-\lambda \phi(t) - g(t - t_0)} \quad ; \quad \phi(t) := \frac{e^{-4at_0} - e^{-4at}}{4a}. \end{aligned} \quad (2.4)$$

Then for $\Xi(\xi)$ we have

$$\partial_\xi^2 \Xi = \frac{\lambda}{D} \xi^2 \Xi, \quad (2.5)$$

which is solvable in terms of the parabolic cylinder function $D_v(z)$ as

$$\Xi(\xi) = c_1 D_{-\frac{1}{2}} \left(x \sqrt{2} \sqrt[4]{\frac{\lambda}{D}} \right) + c_2 D_{-\frac{1}{2}} \left(ix \sqrt{2} \sqrt[4]{\frac{\lambda}{D}} \right). \quad (2.6)$$

Combining $\Xi(\xi)$ and $T(t)$ with the trivially obtained particular solution $V_p(\xi, t) = \frac{p}{g}$ we obtain

$$V(\xi, t) = \left(c_1 D_{-\frac{1}{2}} \left(x \sqrt{2} \sqrt[4]{\frac{\lambda}{D}} \right) + c_2 D_{-\frac{1}{2}} \left(ix \sqrt{2} \sqrt[4]{\frac{\lambda}{D}} \right) \right) e^{-\lambda \phi(t) - g(t - t_0)} + \frac{p}{g}, \quad (2.7)$$

where c_1 and c_2 have been redefined to absorb the factor of T_0 .

3 2D

3.1 Constant Metric

Suppose that h_i are constants. The transformed expression Equation (1.9) becomes

$$\frac{\partial V}{\partial t} = D \left(\frac{1}{h_s^2} e^{-2at} \partial_\xi^2 V + \frac{1}{h_r^2} \partial_r^2 V \right) + p - gV. \quad (3.1)$$

This can be solved analytically using Fourier transforms and working in the Fourier space spanned by $\mathbf{k} = (k_\xi, k_r)^T$. We will use the convention that the Fourier transforms of the function between the two spaces are

$$\begin{aligned}\tilde{V}(\mathbf{k}, t) &= \int_{-\infty}^{\infty} d\xi \int_{-\infty}^{\infty} dr V(\xi, r, t) e^{-i(\mathbf{k}_\xi \xi + \mathbf{k}_r r)}, \\ V(\mathbf{k}, t) &= \iint_{\mathbb{R}^2} d^2\mathbf{k} \tilde{V}(\mathbf{k}, t) e^{i(\mathbf{k}_\xi \xi + \mathbf{k}_r r)}.\end{aligned}\tag{3.2}$$

Moving Equation (3.1) to Fourier space and examining the homogeneous part ($p = 0$) we have

$$\partial_t \tilde{V} = -\left(\frac{D}{h_s^2} e^{-2at} k_\xi^2 + \frac{D}{h_r^2} k_r^2 + g\right) \tilde{V}.\tag{3.3}$$

Integrating the homogeneous equation

$$\begin{aligned}\int_{\tilde{V}_0}^{\tilde{V}} \frac{d\tilde{V}}{\tilde{V}} &= - \int_{t_0}^t dt \left(\frac{D}{h_s^2} e^{-2at} k_\xi^2 + \frac{D}{h_r^2} k_r^2 + g \right) \\ \ln\left(\frac{\tilde{V}}{\tilde{V}_0}\right) &= - \frac{D}{h_s^2} k_\xi^2 \phi(t) - \left(\frac{D}{h_r^2} k_r^2 + g\right)(t - t_0) \quad ; \quad \phi(t) := \frac{e^{-2at_0} - e^{-2at}}{2a} \\ \tilde{V}_c(\mathbf{k}, t) &= \tilde{V}_0(\mathbf{k}, 0) \exp\left(-\frac{D}{h_s^2} k_\xi^2 \phi(t) - \left(\frac{D}{h_r^2} k_r^2 + g\right)(t - t_0)\right).\end{aligned}\tag{3.4}$$

When also including the inhomogeneous source Equation (3.1) in Fourier space becomes

$$\partial_t \tilde{V} = -\left(\frac{D}{h_s^2} e^{-2at} k_\xi^2 + \frac{D}{h_r^2} k_r^2 + g\right) \tilde{V} + p \delta^2(\mathbf{k}),\tag{3.5}$$

which using the complementary function results in the integrating factor

$$\begin{aligned}\tilde{V}(\mathbf{k}, t) &= \tilde{V}_c(\mathbf{k}, t) + p \delta^2(\mathbf{k}) \int_{t_0}^t d\tilde{t} e^{-g(\tilde{t}-t_0)} \\ \tilde{V}(\mathbf{k}, t) &= \tilde{V}_0(\mathbf{k}, 0) \exp\left(-\frac{D}{h_s^2} k_\xi^2 \phi(t) - \left(\frac{D}{h_r^2} k_r^2 + g\right)(t - t_0)\right) + \frac{p}{g} \delta^2(\mathbf{k}) (1 - e^{-g(t-t_0)}),\end{aligned}\tag{3.6}$$

having already used $\mathbf{k} = 0$ in the second term.

Suppose we have a Gaussian packet as our initial condition

$$V(\xi, r, 0) = V_0 \exp\left(-\frac{\xi^2}{4\sigma_\xi^2} - \frac{r^2}{4\sigma_r^2}\right),\tag{3.7}$$

which in Fourier space is

$$\tilde{V}(\mathbf{k}, 0) = V_0 \int_{-\infty}^{\infty} d\xi \int_{-\infty}^{\infty} dr \exp\left(-\frac{\xi^2}{4\sigma_\xi^2} - ik_\xi \xi\right) \exp\left(-\frac{r^2}{4\sigma_r^2} - ik_r r\right),\tag{3.8}$$

then using the Gaussian integral

$$\int_{-\infty}^{\infty} dx \exp(-ax^2 + bx) = e^{\frac{b^2}{4a}} \sqrt{\frac{\pi}{a}} \quad (3.9)$$

this initial condition in Fourier space becomes

$$\tilde{V}(\mathbf{k}, 0) = V_0 e^{-\sigma_\xi^2 k_\xi^2} e^{-\sigma_r^2 k_r^2}. \quad (3.10)$$

Combining this with the general solution in Equation (3.6) (and only focusing on the complementary solution for convenience) we obtain

$$\tilde{V}_c(\mathbf{k}, t) = V_0 e^{-g(t-t_0)} \exp\left(-k_\xi^2 \left(\frac{D}{h_s^2} \phi(t) + \sigma_\xi^2\right)\right) \exp\left(-k_r^2 \left(\frac{D}{h_r^2} (t-t_0) + \sigma_r^2\right)\right). \quad (3.11)$$

To move back to position space we use the Fourier transform in Equation (3.2) and the Gaussian integral to obtain

$$\begin{aligned} V(\xi, r, t) &= V_0 \tau^{-2} e^{-g(t-t_0)} \int_{-\infty}^{\infty} dk_\xi \exp\left(-k_\xi^2 \left(\frac{D}{h_s^2} \phi(t) + \sigma_\xi^2\right) + ik_\xi \xi\right) \\ &\quad \cdot \int_{-\infty}^{\infty} dk_r \exp\left(-k_r^2 \left(\frac{D}{h_r^2} (t-t_0) + \sigma_r^2\right) + ik_r r\right) \\ &\quad + \frac{p}{g} \iint_{\mathbb{R}^2} d^2 \mathbf{k} \delta^2(\mathbf{k}) (1 - e^{-g(t-t_0)}) \\ &= \frac{V_0 e^{-g(t-t_0)}}{2\tau \sqrt{\left(\frac{D}{h_s^2} \phi(t) + \sigma_\xi^2\right) \left(\frac{D}{h_r^2} (t-t_0) + \sigma_r^2\right)}} \\ &\quad \cdot \exp\left(-\frac{\xi^2}{4\left(\frac{D}{h_s^2} \phi(t) + \sigma_\xi^2\right)}\right) \exp\left(-\frac{r^2}{4\left(\frac{D}{h_r^2} (t-t_0) + \sigma_r^2\right)}\right). \end{aligned} \quad (3.12)$$

3.2 Small a

For the parabolic map Equation (1.9) becomes

$$\partial_t V = \frac{D}{\rho^2} \left(e^{-2at} \partial_\xi^2 V + \partial_r^2 V \right), \quad (3.13)$$

where $\rho := \sqrt{s^2 + r^2} = \sqrt{\xi^2 e^{2at} + r^2}$. To use perturbation theory we say that $a \ll 1$ and $V(\xi, r, t) = V_0(\xi, r, t) + aV_1(\xi, r, t)$. Expanding in a first of all we obtain

$$\begin{aligned} \partial_t V &= \frac{D}{\xi^2 + r^2 + 2at\xi^2} \left((1 - 2at) \partial_\xi^2 V + \partial_r^2 V \right) + p - gV \\ \left((\xi^2 + r^2) (\partial_t + g) - D(\partial_\xi^2 + \partial_r^2) + a(2Dt\partial_\xi^2 + \partial_t + g) \right) V &= (\xi^2 + r^2 + 2at\xi^2)p. \end{aligned} \quad (3.14)$$

3.2.1 $\mathcal{O}(a^0)$

At $\mathcal{O}(a^0)$ we are examining a system without growth so both the ξ and r directions are treated on equal footing, that is we have an $O(2)$ symmetry group. The resulting expression is

$$\left((\xi^2 + r^2)(\partial_t + g) - D(\partial_\xi^2 + \partial_r^2)\right)V_0 = (\xi^2 + r^2)p. \quad (3.15)$$

Due to the symmetry now present it is natural to use polar coordinates (λ, θ) with $\lambda^2 \equiv \xi^2 + r^2$. Using the Cartesian Laplacian in polar coordinates

$$\partial_\xi^2 + \partial_r^2 = \partial_\lambda^2 + \frac{1}{\lambda}\partial_\lambda + \frac{1}{\lambda^2}\partial_\theta^2 \quad (3.16)$$

we arrive at

$$\left(\lambda^2(\partial_t + g) - D(\partial_\lambda^2 + \frac{1}{\lambda}\partial_\lambda + \frac{1}{\lambda^2}\partial_\theta^2)\right)V_0 = \lambda^2 p. \quad (3.17)$$

Then focusing on the homogeneous LHS and using the separation of variables $V_0(\lambda, \theta, t) = T(t)\Lambda(\lambda)\Theta(\theta)$ we obtain

$$\lambda^2 \frac{\partial_t T}{T} + \lambda^2 g - D\left(\frac{\partial_\lambda^2 \Lambda}{\Lambda} + \frac{\partial_\lambda \Lambda}{\lambda \Lambda} + \frac{\partial_\theta^2 \Theta}{\lambda^2 \Theta}\right) = 0. \quad (3.18)$$

There is only one term that depends on Θ and so all the θ dependence must be a constant. We choose

$$\partial_\theta^2 \Theta = -m^2 \Theta \quad (3.19)$$

where m is some constant. This possesses oscillatory solutions of the form $\Theta(\theta) = A_\pm e^{\pm im\theta}$. One can do the same with the time dependence, although in this case we can include the degradation term to and select

$$\frac{\partial_t T}{T} + g = -\mu, \quad (3.20)$$

for some arbitrary constant μ . This can be integrated to find

$$T(t) = B e^{-(g+\mu)t}. \quad (3.21)$$

Substituting the constants into Equation (3.18) we obtain

$$\lambda^2 \frac{\mu}{D} + \frac{\partial_\lambda^2 \Lambda}{\Lambda} + \frac{\partial_\lambda \Lambda}{\lambda \Lambda} - \frac{m^2}{\lambda^2} = 0, \quad (3.22)$$

which has solutions in terms of Bessel functions of the first kind $J_n(z)$ and gamma functions $\Gamma(x)$ as

$$\Lambda(\lambda) = c_- J_{-\frac{m}{2}}\left(\frac{\lambda^2}{2}\sqrt{\frac{\mu}{D}}\right)\Gamma\left(1 - \frac{m}{2}\right) + c_+ J_{\frac{m}{2}}\left(\frac{\lambda^2}{2}\sqrt{\frac{\mu}{D}}\right)\Gamma\left(1 + \frac{m}{2}\right). \quad (3.23)$$

Combining everything we obtain

$$\begin{aligned} V_0^c(\lambda, \theta, t) = & e^{-(g+\mu)t} (e^{im\theta} + A e^{-im\theta}) \\ & \cdot \left(c_- J_{-\frac{m}{2}}\left(\frac{\lambda^2}{2}\sqrt{\frac{\mu}{D}}\right)\Gamma\left(1 - \frac{m}{2}\right) + c_+ J_{\frac{m}{2}}\left(\frac{\lambda^2}{2}\sqrt{\frac{\mu}{D}}\right)\Gamma\left(1 + \frac{m}{2}\right) \right) \end{aligned} \quad (3.24)$$

for the complementary solution, having rescaled constants set by boundary conditions.

Including the inhomogeneous term $\lambda^2 p$ we form the ansatz for the particular solution $V_0^p = \alpha\lambda^2 + \beta\lambda + \gamma$. Substituting this into Equation (3.17) we obtain

$$\lambda^2 g(\alpha\lambda^2 + \beta\lambda + \gamma) - D\left(2\alpha + 2\alpha + \frac{\beta}{\lambda}\right) = \lambda^2 p, \quad (3.25)$$

which implies $\alpha = 0$, $\beta = 0$, $\gamma = \frac{p}{g}$. Therefore the full $\mathcal{O}(a^0)$ solution is

$$V_0(\lambda, \theta, t) = e^{-(g+\mu)t} (e^{im\theta} + Ae^{-im\theta}) \cdot \left(c_- J_{-\frac{m}{2}} \left(\frac{\lambda^2}{2} \sqrt{\frac{\mu}{D}} \right) \Gamma\left(1 - \frac{m}{2}\right) + c_+ J_{\frac{m}{2}} \left(\frac{\lambda^2}{2} \sqrt{\frac{\mu}{D}} \right) \Gamma\left(1 + \frac{m}{2}\right) \right) + \frac{p}{g}. \quad (3.26)$$

3.2.2 $\mathcal{O}(a^1)$

If we say that the $\mathcal{O}(a^0)$ homogeneous dynamics are encapsulated by $\mathcal{L} = (\xi^2 + r^2)(\partial_t + g) - D(\partial_\xi^2 + \partial_r^2)$ then the $\mathcal{O}(a^1)$ expression is

$$\mathcal{L}V_1 = 2t\xi^2 p - (2Dt\partial_\xi^2 + \partial_t + g)V_0, \quad (3.27)$$

where the RHS inhomogeneous part is responsible for the breaking of the $O(2)$ symmetry. Clearly, $\partial_t V_0 = -(g + \mu)V_0^c$. Furthermore, using the Jacobian from Cartesian to Polar coordinates

$$J_{\xi r}^{\lambda\theta} = \begin{pmatrix} \frac{\xi}{\lambda} & \frac{r}{\lambda} \\ -\frac{r}{\lambda^2} & \frac{\xi}{\lambda^2} \end{pmatrix} \quad (3.28)$$

and the fact that the derivative of the Bessel function of the first kind with respect to its argument can be expressed as

$$\partial_z J_n(z) = \frac{1}{2}(J_{n-1}(z) - J_{n+1}(z)) \quad (3.29)$$

we can find

$$\begin{aligned} \partial_\xi e^{im\theta} &= \frac{\partial\theta}{\partial\xi} im e^{im\theta} = -\frac{imr}{\lambda^2} e^{im\theta} \\ \partial_\xi^2 e^{im\theta} &= -\frac{m^2 r^2}{\lambda^4} e^{im\theta} + 2imr\lambda^{-3} \frac{\partial\lambda}{\partial\xi} e^{im\theta} = \frac{2imr\xi - m^2 r^2}{\lambda^4} e^{im\theta} \\ \partial_\xi J_n(L\lambda^2) &= (J_{n-1}(L\lambda^2) - J_{n+1}(L\lambda^2))\lambda L \frac{\partial\lambda}{\partial\xi} \\ &= (J_{n-1}(L\lambda^2) - J_{n+1}(L\lambda^2))\xi L \\ \partial_\xi^2 J_n(L\lambda^2) &= \left((J_{n-2}(L\lambda^2) - 2J_n(L\lambda^2)) + J_{n+2}(L\lambda^2) \right) \xi^2 L^2 \\ &\quad + (J_{n-1}(L\lambda^2) - J_{n+1}(L\lambda^2))L, \end{aligned} \quad (3.30)$$

where $L := \frac{1}{2}\sqrt{\frac{\mu}{D}}$. So we then have

$$\begin{aligned}
e^{(g+\mu)t}\partial_\xi^2 V_0 = & \lambda^{-4}((2imr\xi - m^2r^2)e^{im\theta} + A(-2imr\xi - m^2r^2)e^{-im\theta}) \\
& \cdot \left(c_- J_{-\frac{m}{2}}(L\lambda^2)\Gamma\left(1 - \frac{m}{2}\right) + c_+ J_{\frac{m}{2}}(L\lambda^2)\Gamma\left(1 + \frac{m}{2}\right) \right) \\
& + (e^{im\theta} + Ae^{-im\theta}) \left(c_- \left[\left((J_{-\frac{m}{2}-2}(L\lambda^2) - 2J_{-\frac{m}{2}}(L\lambda^2)) + J_{-\frac{m}{2}+2}(L\lambda^2) \right) \xi^2 L^2 \right. \right. \\
& + \left. \left. (J_{-\frac{m}{2}-1}(L\lambda^2) - J_{-\frac{m}{2}+1}(L\lambda^2))L \right] \Gamma\left(1 - \frac{m}{2}\right) \right. \\
& + c_+ \left[\left((J_{\frac{m}{2}-2}(L\lambda^2) - 2J_{\frac{m}{2}}(L\lambda^2)) + J_{\frac{m}{2}+2}(L\lambda^2) \right) \xi^2 L^2 \right. \\
& + \left. \left. (J_{\frac{m}{2}-1}(L\lambda^2) - J_{\frac{m}{2}+1}(L\lambda^2))L \right] \Gamma\left(1 + \frac{m}{2}\right) \right). \tag{3.31}
\end{aligned}$$

The full expression now takes the form

$$\mathcal{L}V_1 = 2t\xi^2 p + \mu V_0^c - p - 2Dt\partial_\xi^2 V_0. \tag{3.32}$$